

A fuzzy based approach for the selection of goals in goal oriented requirements elicitation process

Mohd Sadiq · S. K. Jain

Received: 20 January 2014 / Revised: 24 March 2014 / Published online: 1 June 2014

© The Society for Reliability Engineering, Quality and Operations Management (SREQOM), India and The Division of Operation and Maintenance, Lulea University of Technology, Sweden 2014

Abstract Goal selection in goal oriented requirements elicitation process (GOREP) like KAOS, i*, AGORA etc., is a multi-criteria decision making problem. In literature, we identify that existing GOREP do not support to choose and adopt a goal out of the alternatives of the decomposed goals when multi-criteria decision making approaches are used and stakeholders' preferences are in the form of linguistic variables. Therefore, this paper presents a fuzzy based approach to address the above issue by using α -level weighted F-preference relation in group decision making process. Finally, the utilization of the proposed approach is demonstrated with the help of an example.

Keywords Goal oriented requirements engineering · Requirements elicitation · Fuzzy logic · Decision making approach

1 Introduction

Goals are high level objective of an organization and are the basis of goal oriented requirements engineering (GORE) (Lamsweerde 2001). In requirements elicitation process, requirements are identified by group of stakeholders on the basis of domain knowledge, business objective, and product vision (Sadiq and Jain 2012; Nuseibeh and Easterbrook 2002; Potts 1995). There are

various methods for requirements elicitation like Goal Oriented Requirements Elicitation process (Kavakli 2002; Kaiya et al. 2002; Anwer and Ikram 2006; Yu 1997; Zickert 2010), Quality Function Deployment (QFD) (Chan and Wu 2007; Pai 2006) etc. In goal oriented requirements elicitation process, high level objective of an organization are decomposed into sub-goals/requirements to construct an AND/OR graph. This graph is used to find out the functional and non functional requirements of software (Chung et al. 2000).

In an AND/OR graph, there are two types of decomposition of a goal into sub-goals that use *AND* and *OR* logical connectives. An *AND* decomposition implies that a parent goal can be achieved/satisfied if all its sub-goals are achieved/satisfied. Similarly, an *OR* decomposition achieves a parent goal if its any sub-goal is achieved (Kaiya et al. 2005).

QFD is a requirements gathering tool for developing new or improved products to increase customers satisfaction by integrating design engineering, marketing, manufacturing and other related activities of an organization (Karsak 2004). QFD uses four matrixes namely product planning, part planning, process planning, and production planning to organize and relate pieces of data to each other (Chan and Wu 2007). These matrixes are combined to form a basis tool of QFD, called House of quality (HoQ). QFD has been widely applied in various areas like transportation and communication, electronics and electrical utilities, food processing, textile, and construction etc. (Prasad 1998; Pai 2006).

There are various fundamental challenges in RE like producing well formulated requirements, managing the constant flow of requirements, requirements volatility, requirements traceability and interdependence, selecting the right process, and software release planning (Karlsson

M. Sadiq (✉) · S. K. Jain
Computer Engineering Department, National Institute of
Technology, Kurukshetra 136119, Haryana, India
e-mail: sadiq.jmi@gmail.com

S. K. Jain
e-mail: skj_nith@yahoo.com

et al. 2007). Karlsson et al. (2007) pointed out that among various RE challenges; software release planning is a major challenge in RE. It deals with the selection and prioritization of goals for the next release of software product according to the need of stakeholders. Software companies are facing challenges in order to determine the set of goals for the next release of software product. A system may have thousand of goal/requirements proposed by stakeholders; and it is difficult to select these goals/requirements that add most value to the software product (Akker et al. 2005). For this reason, we mainly focus on the selection of goals in this work.

Selection of a goal from a set of goals, on the basis of many criteria, creates a multi-criteria decision making problems (MCDM). Since different stakeholders participate in requirements elicitation process, therefore, they have different preferences for the same goal because of their experience, knowledge, and understanding levels (Wang and Xiong 2011; Lee 2005). Key elements in human thinking are not numbers, but linguistic terms or fuzzy set levels (Lin et al. 2007). The values given by these stakeholders to a goal are often qualitatively described or imprecisely measured because human judgements including preferences are often vague and cannot estimate his preference with exact numerical values (Zadeh 1975, 1965). Fuzzy logic can handle imprecise and linguistic variables in a mathematically well defined way, for example, expressions such as: “high performance”, and low cost are imprecise (Lin et al. 2007; Lee 2005; Hwang and Yoon 1981). In literature, fuzzy based decision making methods have gained acceptance over the last decade in different area of science and engineering, for example, technology transfer selection in biotechnology (Chang and Chen 1994), supplier selection problem (Li et al. 2007), selection of object oriented simulation software for production system analysis (Cochran and Chen 2005), and a non functional requirements tradeoffs model in trustworthy software (Zhu et al. 2012), software component selection (Yamamoto and Saeki 2007) etc. In 2013, we present a fuzzy based approach for requirements prioritization in goal oriented requirements elicitation process (Sadiq and Jain 2013). The contribution of this paper was to extend the attributed goal oriented requirements analysis (AGORA) method by applying the fuzzy based approach and binary sort tree method for the selection and prioritization of goals in goal oriented requirements elicitation process. We demonstrate the proposed method by considering the ten requirements (Sadiq and Jain 2013).

In literature, we identify that goal oriented requirements elicitation processes like knowledge acquisition for automated specification (KAOS) (Zickert 2010), i* (Yu 1997), AGORA (Kaiya et al. 2002) etc. do not support to choose

and adopt a goal out of the alternatives of the decomposed goals when MCDM approaches are used; and stakeholders' preferences are in the form of linguistic variables (Anton 1996; Akker et al. 2005; Kaiya et al. 2002; Kaiya et al. 2005; Kavakli 2002; Oshiro et al. 2003; Sadiq and Jain 2013; Shibaoka et al. 2007; Yu E S 1997; Zickert 2010). Therefore, to address this issue, we present a fuzzy based approach by combining an AND/OR goal graph and α -level weighted F-preference relation proposed by Li (1999) in group decision making process for the selection of goals when MCDM approach is used in fuzzy environment and multiple stakeholders participate in requirements elicitation method.

This paper is organized as follows: In Sect. 2, basic concepts of fuzzy sets, linguistic variable, fuzzy triangular numbers, and fuzzy preference relations are reviewed. In Sect. 3, we present fuzzy approach for goal selection. An example is provided in Sect. 4 to show how the proposed approach works under fuzzy group decision making. Finally, conclusions are drawn in Sect. 5.

2 Fuzzy set theory

In this section, we briefly review the basic concepts of fuzzy sets, linguistic variable, fuzzy triangular numbers, and fuzzy preference relation. The fuzzy set, originally proposed by Zadeh (1965), is defined as follows: In a universe of discourse U_x , a fuzzy subset A of U_x is characterized by a membership function $f_A(x)$, where $f_A : U_x \rightarrow [0, 1]$ and the membership function associates with each member of x of U_x a number of $f_A(x)$ in the interval $[0,1]$, representing the grade of membership of x in A . Linguistic variables are variables whose values are words or sentences in a natural or artificial language (Zadeh 1965). For example, poor is a linguistic variable if its values are assumed to be the fuzzy variables labelled very poor, poor, fair, good, and very good; rather than the numbers 0,1,2,3 etc.

There are several formats of fuzzy numbers, such as Triangular, Trapezoidal, Gaussian, or Sigmoid that can be used in decision making processes. In practical applications, *triangular fuzzy numbers (TFNs)* are widely used to represent the approximate value range of linguistic variables (Mahmood et al. 2013). In the proposed method we adopt TFNs because of their simplicity in both concepts and computation (Li 1999; Liu 2010). TFNs can be defined as follows:

Let $\mathbf{R}$ is the real line, which is viewed as a universal set of all fuzzy sub-sets. A triangular fuzzy number A is normal, convex fuzzy subset of $\mathbf{R}$, with a piece wise linear relationship function μ_A , defined by:

$$\mu_A(x) = \begin{cases} \frac{(x-a)}{(b-a)}, & a \leq x \leq b, \\ \frac{(c-x)}{(c-b)}, & b \leq x \leq c, \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

The TFNs can be denoted by $A = (a, b, c)$ as depicted in Fig. 1. The parameters a , b , and c respectively, indicate the smallest possible value, the most promising value, and the largest possible value that describe a fuzzy event. There are several operations that can be performed on triangular fuzzy numbers (TFN) like addition, subtraction, inverse etc.

Let $A_1 = (a_1, b_1, c_1)$ and $A_2 = (a_2, b_2, c_2)$ then:

$$\text{Addition : } A_1 \oplus A_2 = (a_1 + a_2, b_1 + b_2, c_1 + c_2) \quad (2)$$

$$\text{Subtraction : } A_1 \ominus A_2 = (a_1 - c_2, b_1 - b_2, c_1 - a_2) \quad (3)$$

$$\text{Multiplication : } A_1 \odot A_2 = (a_1.a_2, b_1.b_2, c_1.c_2) \quad (4)$$

$$\text{Inverse : } (A_1)^{-1} = (1/c_1, 1/b_1, 1/a_1) \quad (5)$$

$$\text{Negation of } A_1 = (-c_1, -b_1, -a_1) \quad (6)$$

$$\text{Division : } A_1/A_2 = (a_1/c_2, b_1/b_2, c_1/a_2) \quad (7)$$

Preference relation is a useful tool for representation of information used in decision making problems. It is used when we want to aggregate expert's preferences into group preferences. A fuzzy preference relation P on R is a fuzzy subset of $R \times R$ with membership function $f_P(A, B \forall A, B \subseteq R)$, here $f_P(A, B)$ represents the degree of preference of A over B (Li 1999; Lee 2005; Nakamura 1986):

1. P is reciprocal iff $f_P(A, B) = 1 - f_P(B, A)$, $A, B \subseteq R$.
2. P is transitive iff $f_P(A, B) \geq 1/2$ and $f_P(B, C) \geq 1/2 \rightarrow f_P(A, C) \geq 1/2 \forall A, B, C \subseteq R$.
3. P is a fuzzy total ordering iff p is reciprocal, transitive, and comparable.

3 A fuzzy based approach for goal selection

This section presents a fuzzy based approach for goal selection by combining an AND/OR goal graph and α -level

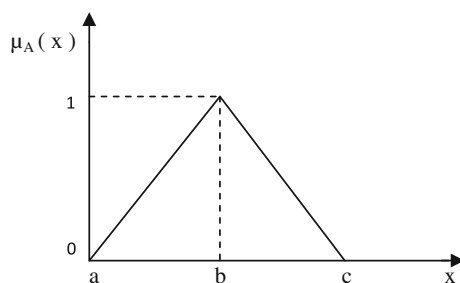


Fig. 1 The membership function of a TFN $A = (a, b, c)$

weighted F-preference relation proposed by Li (1999) when MCDM approach is used in fuzzy environment and multiple stakeholders participate in requirements analysis method. The proposed method is presented in Fig. 2.

Step 1: Identify stakeholders and their goals: The first step of the proposed method is to identify stakeholders and their goals. Primary stakeholders include those who are central to any project initiative, i.e., beneficiaries, financial, politicians, sponsors, and decision maker. Secondary stakeholders include developers, experts, operators etc. Requirements analyst identifies the high level objective of the primary stakeholders and may use the following elicitation techniques to identify the goal: (i) interview (ii) asking 'why', 'how', and 'how-else' questions (iii) analysis of the existing documents etc. The 'why' questions are used to identify the objective of the stakeholders; how questions are used to identify lower level goals; and the how-else questions are used to find the alternatives to satisfy the higher level goals (Lamsweerde 2001).

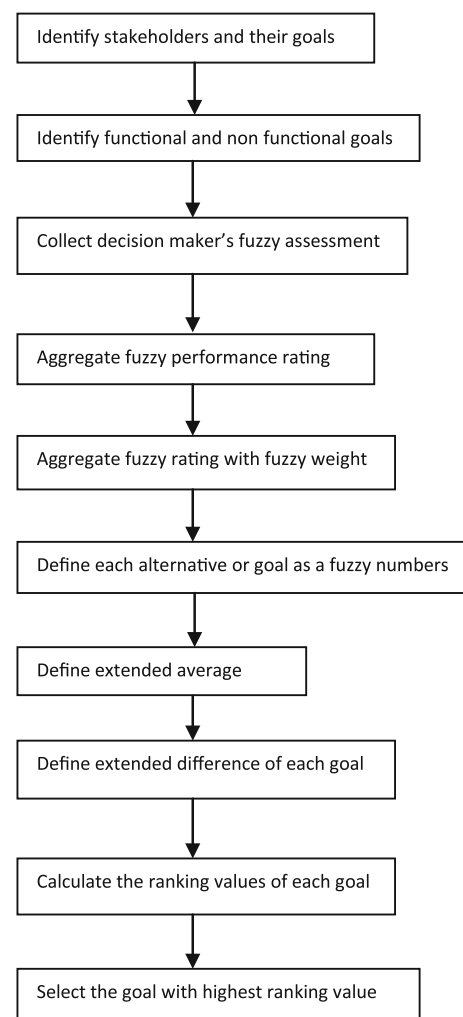


Fig. 2 Proposed method for goal selection in GOREP

Step 2: Identify functional and non-functional goal:

In this step, identified goal is decomposed into sub goals to construct an AND/OR graph. This graph is used to identify the functional and non functional goal. In AND decomposition, if all of the sub goals are achieved, their parent goals can be achieved or satisfied. On the other hand side, in OR decomposition, the achievement of at least one-sub goals leads to the achievement of its parent goal. Let the decomposed goals are represented by $G_1, G_2, \dots, G_m$.

Step 3: Collect decision maker's fuzzy assessment:

Let M decision makers (DM) are participating during requirements analysis to select the goal from the set of goals. Then, we collect the experts' fuzzy assessments and express their opinions on the importance of each sub-goal. DM may have different view, originating from different roles and their knowledge, so they may give different preferences in the form of linguistic variables such as very low, low, high, very high etc.

Step 4: Aggregate fuzzy performance rating through all decision maker by means of extended addition and scalar multiplication to form a comprehensive performance matrix P , in which performance rating $p_{ij} = (1/n) \odot p_{ij}^1 \oplus p_{ij}^2 \oplus \dots \oplus p_{ij}^n$ is a triangular fuzzy number of the form:

$$(p_{1ij}, p_{2ij}, p_{3ij}) = \left(\frac{1}{n} \sum_{k=1}^n p_{1ij}^k, \frac{1}{n} \sum_{k=1}^n p_{2ij}^k, \frac{1}{n} \sum_{k=1}^n p_{3ij}^k \right)$$

Now calculate the fuzzy weight through all decision makers by means of extended addition and scalar multiplication to form a comprehensive weight vector W (Li 1999).

Step 5: Aggregate fuzzy ratings with fuzzy weights by means of extended multiplication to form a weighted, comprehensive decision matrix D , in which $d_{ij} = p_{ij} \odot_j$ is a fuzzy number with parabolic membership functions in the form of (Li 1999):

$$(\lambda_{1ij}, \lambda_{2ij}, \lambda_{3ij}/d_{ij}/\Delta_{1ij}, \Delta_{2ij}, \Delta_{3ij}),$$

Where

$$\lambda_{1ij} = (w_{2j} - w_{1j})(p_{2ij} - p_{1ij}),$$

$$\lambda_{2ij} = w_{1j}(p_{2ij} - p_{1ij}) + p_{1ij}(w_{2j} - w_{1j}),$$

$$\lambda_{3ij} = w_{1j}p_{1ij},$$

$$\Delta_{1ij} = (w_{3j} - w_{2j})(p_{3ij} - p_{2ij}),$$

$$\Delta_{2ij} = w_{3j}(p_{3ij} - p_{2ij}) + p_{3ij}(w_{3j} - w_{2j}),$$

$$\Delta_{3ij} = w_{3ij}p_{3ij}, \text{ and}$$

$$d_{ij} = w_{2j}p_{2ij}$$

Step 6: Define each alternative (sub-goals) as a fuzzy number A_i , $i = 1, 2, \dots, m$ by means of extended

addition and scalar multiplication through the criteria (C), i.e.,

$A_i = 1/C \odot (d_{i1} \oplus d_{i2} \oplus \dots \oplus d_{iC})$ with parabolic membership function in the form of (Li 1999):

$$(\lambda_{1i}, \lambda_{2i}, \lambda_{3i}/A_i/\Delta_{1i}, \Delta_{2i}, \Delta_{3i}),$$

where

$$\lambda_{1i} = \frac{1}{C} \sum_{j=1}^C \lambda_{1ij}, \quad I = 1, 2, 3,$$

$$\Delta_{1i} = \frac{1}{C} \sum_{j=1}^C \Delta_{1ij}, \quad I = 1, 2, 3, \text{ and}$$

$$A_i \frac{1}{C} \sum_{j=1}^C d_{ij}$$

Step 7: Define extended average $\hat{A}$ by means of extended addition and scalar multiplication through all alternatives (sub-goals), i.e., $A_{1i} = 1/m \odot (G_1 \oplus G_2 \oplus \dots \oplus G_m)$ with parabolic membership function in the form of (Li 1999):

$$(\lambda_1, \lambda_2, \lambda_3/\hat{A}/\Delta_1, \Delta_2, \Delta_3),$$

where

$$\lambda_1 = \frac{1}{m} \sum_{j=1}^m \lambda_{1i}, \quad I = 1, 2, 3,$$

$$\Delta_1 = \frac{1}{m} \sum_{j=1}^m \Delta_{1i}, \quad I = 1, 2, 3, \text{ and}$$

$$\hat{A} = \frac{1}{m} \sum_{j=1}^m A_i$$

Step 8: Define the extended difference, $A_i \ominus \hat{A}$, for each $A_i \in R$, with parabolic membership function in the form of (Li 1999):

$$\left((\lambda_{1i} - \Delta_1), (\lambda_{2i} + \Delta_2), (\lambda_{3i} - \Delta_3)/A_i - \hat{A}/(\Delta_{1i} - \lambda_1), \right. \\ \left. (-\Delta_{2i} - \lambda_2), (\Delta_{3i} - \lambda_3) \right).$$

Step 9: Calculate the ranking values r_i for each alternative (sub-goals) A_i by means of F-preference relation R (Li 1999):

if $(\lambda_{3i} - \Delta_3) < 0$, $(\Delta_{3i} - \lambda_3) \geq 0$, $A_i \geq \hat{A}$;

then $r_i = \mu_R(A_i \ominus \hat{A}, 0) = \prod^+ / (\prod^+ + \prod^-)$,

else if $(\lambda_{3i} - \Delta_3) \leq 0$, $(\Delta_{3i} - \lambda_3) > 0$, $A_i \leq \hat{A}$;

then $r_i = \mu_R(A_i \ominus \hat{A}, 0) = \Psi^+ / (\Psi^+ + \Psi^-)$,

else if $(\lambda_{3i} - \Delta_3) = 0$, $(\Delta_{3i} - \lambda_3) = 0$, $A_i = \hat{A}$.

then $r_i = \mu_R(A_i \ominus \hat{A}, 0) = 0.5$,

else if $(\lambda_{3i} - \Delta_3) \geq 0, (\Delta_{3i} - \lambda_3) > 0, A_i \geq \hat{A}$.

then $r_{i=} \mu_R (A_i \ominus \hat{A}, 0) = 1$,

else if $(\lambda_{3i} - \Delta_3) < 0, (\Delta_{3i} - \lambda_3) \leq 0, A_i \leq \hat{A}$.

then $r_{i=} \mu_R (A_i \ominus \hat{A}, 0) = 0$.

where

$$\begin{aligned} \Pi^+ &= [1/4(\Delta_{1i} - \lambda_1) - 1/3(\Delta_{2i} + \lambda_2) + 1/2(\Delta_{3i} - \lambda_3)] \\ &\quad + [1/4(\lambda_{1i} - \Delta_1)(1 - \tilde{r}^4) + 1/3(\lambda_{2i} + \Delta_2)(1 - \tilde{r}^3) \\ &\quad + 1/2(\lambda_{3i} - \Delta_3)(1 - \tilde{r}^2)]; \\ \Pi^- &= (1/4(\lambda_{1i} - \Delta_1)\tilde{r}^4 + 1/3(\lambda_{2i} + \Delta_2)\tilde{r}^3 \\ &\quad + 1/2(\lambda_{3i} - \Delta_3)\tilde{r}^2)]; \\ \tilde{r} &= [- (\lambda_{2i} + \Delta_2) + \text{sqrt}\{(\lambda_{2i} + \Delta_2)^2 - 4(\lambda_{1i} - \Delta_1) \\ &\quad \times (\lambda_{3i} - \Delta_3)\}]/[2(\lambda_{1i} - \Delta_1)]; \\ \Psi^+ &= 1/4(\Delta_{1i} - \lambda_1)\tilde{r}^4 + 1/3(-\Delta_{2i} - \lambda_2)\tilde{r}^3 \\ &\quad + 1/2(\Delta_{3i} - \lambda_3)\tilde{r}^2; \\ \Psi^- &= - [1/4(\lambda_{1i} - \Delta_1) + 1/3(\lambda_{2i} + \Delta_2) \\ &\quad + 1/2(\lambda_{3i} - \Delta_3)] - [1/4(\Delta_{1i} - \lambda_1)(1 - \tilde{r}^4) \\ &\quad - 1/3(\Delta_{2i} + \lambda_2)(1 - \tilde{r}^3) + 1/2(\Delta_{3i} - \lambda_3)(1 - \tilde{r}^2)]; \\ \tilde{r} &= [(\Delta_{2i} + \lambda_2) - \text{sqrt}\{(-\Delta_{2i} - \lambda_2)^2 - 4(\Delta_{1i} - \lambda_1) \\ &\quad (\Delta_{3i} - \lambda_3)\}]/[2(\Delta_{1i} - \lambda_1)]; \end{aligned}$$

$\tilde{r} = [(\Delta_{2i} + \lambda_2) - \text{sqrt}\{(-\Delta_{2i} - \lambda_2)^2 - 4(\Delta_{1i} - \lambda_1) (\Delta_{3i} - \lambda_3)\}]/[2(\Delta_{1i} - \lambda_1)]$; where “sqrt” is a square root function?

Step 10: Select the goal with the maximum ranking values: Arrange the values in ascending order and select the goal with the maximum ranking value as the best alternative.

4 An example

The objective of this section is to apply the proposed method for the selection of goals for institute examination system. This system is used to provide the facility to submit online examination form and generate the results of the

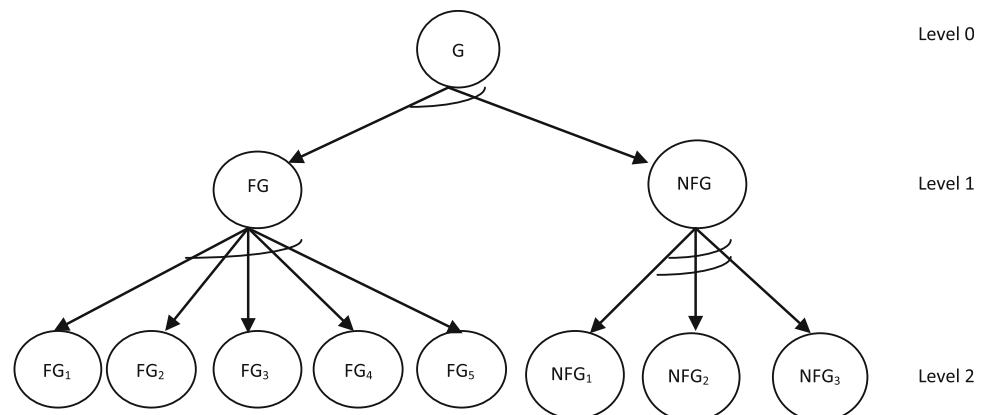
students; and display the news related to examinations. After submitting the examination form, the system will generate the examination fee receipt and students will be able to take the printout of that receipt. The system will also generate the patterns of the sitting arrangement. Goal selection plays a key role in requirements elicitation process to decide which requirements to implement first according to the budget, time, cost, customer expectations etc.

Step 1: To illustrate the proposed approach, we assume that ten stakeholders i.e., $S_1, S_2, S_3, S_4, S_5, S_6, S_7, S_8, S_9$, and S_{10} ; and two decision makers (DM), i.e. DM_1 , and DM_2 , are participating in group requirements elicitation and analysis process.

Let S_1 is the primary stakeholder and its objective is to develop software for institute examination system. S_1 is also the beneficiary and sponsor of the system, who wants to develop a system. S_2 is the secondary stakeholder and is responsible for the identification of functional goal (FG) and to deal with the decisions that are related to the organization goals and objective. The objective of stakeholder S_3 is to elicit the non-functional goals and also to take the decisions related to the organizations goals and objective. Stakeholder S_4 is the in-charge of the system throughout all the life cycle phases. Stakeholder S_5 is requirements analyst and developer; and is directly involved in information system development. S_6 is operator and deals with assuring effectiveness when performing operation within the organization. S_7 is both primary and secondary stakeholders and will act as an expert and the decision maker. S_8 is the regulators and is responsible for quality, security and cost or other aspects of the system. S_9 and S_{10} are the financers; and they benefit indirectly from the system, obtaining financial rewards and are selected on the basis of geographical location.

Step 2: Requirements elicitation process is used to identify the requirements of software according to the need of stakeholders (Sadiq and Shahid 2009). Let us assume

Fig. 3 AND/OR graph



that during group requirements elicitation process S_5 construct an AND/OR graph by decomposing and refining the objective of S_1 into five functional goals (FG), i.e., student module (FG_1), teachers module (FG_2), controller of examination module (FG_3), administrative module (FG_4), and online conduct of examination module (FG_5); and three non-functional goals (NFG) or criteria (Fukushima and Yamada 2010; Chung et al. 2000), i.e., effort (NFG_1), cost (NFG_2), risk (NFG_3) as shown in Fig. 3. In this graph high level objective is represented by G. This goal is decomposed into two sub-goals, i.e., FG and NFG. There is an AND decomposition between FG and NFG at level 1. At level 2, there is an AND decomposition among FG_1 , FG_2 , FG_3 , FG_4 , and FG_5 ; there is an OR decomposition among NFG_1 , NFG_2 , and NFG_3 .

Step 3: Once, we have identified the functional and non-functional goals then the next step is to collect the DM fuzzy assessment (Step 3). Fuzzy assessment of two DM is given in Tables 1 and 2. Fuzzy importance weight of two DMs, i.e., DM_1 and DM_2 is given in Table 3.

After applying Step 4, we get the values of the comprehensive performances and comprehensive weight, of each goal, as shown in Tables 4 and 5 respectively.

Table 1 Fuzzy assessment by first DM

Sub-goals	Non-functional goal (NFG)		
	NFG_1	NFG_2	NFG_3
FG_1	(2,4,5)	(3,4,5)	(4,6,7)
FG_2	(2,3,6)	(2,5,8)	(3,4,7)
FG_3	(1,2,2)	(3,5,7)	(3,6,7)
FG_4	(5,6,7)	(2,3,6)	(3,4,5)
FG_5	(1,2,3)	(2,3,4)	(2,4,6)

Table 2 Fuzzy assessment by second DM

Sub-goals	Non-functional goal (NFG)		
	NFG_1	NFG_2	NFG_3
FG_1	(5,6,6)	(2,5,6)	(5,7,8)
FG_2	(3,4,6)	(2,2,3)	(3,4,5)
FG_3	(4,6,6)	(5,6,7)	(2,3,3)
FG_4	(1,2,2)	(3,2,2)	(1,2,3)
FG_5	(2,3,2)	(3,4,5)	(2,3,4)

Table 3 Fuzzy importance weight

NFG	First DM	Second DM
NFG_1	(0.2,0.3,0.6)	(0.3,0.6,0.7)
NFG_2	(0.3,0.4,0.6)	(0.2,0.4,0.5)
NFG_3	(0.1,0.2,0.3)	(0.1,0.3,0.4)

Table 4 Comprehensive performance

Sub-goals	Non-functional goal (NFG)		
	NFG_1	NFG_2	NFG_3
FG_1	(3.5,5,5.5)	(2.5,4.5,5.5)	(4.5,6.5,7.5)
FG_2	(2.5,3.5,6)	(2,3.5,5.5)	(3,4,6)
FG_3	(2.5,4,4)	(4,5,5,7)	(2.5,4.5,5)
FG_4	(3,4,4.5)	(2.5,4,4)	(2,3,4)
FG_5	(1.5,2.5,2.5)	(2.5,3.5,4.5)	(2,3.5,5)

Table 5 Comprehensive weight

NFG_1	NFG_2	NFG_3
(0.25,0.45,0.65)	(0.25,0.4,0.55)	(0.1,0.25,0.35)

Table 6 Quadratic membership function

Sub-goals	Non-functional goal (NFG)		
	NFG_1	NFG_2	NFG_3
FG_1	2.25_{QF}	1.8_{QF}	1.625_{QF}
FG_2	1.575_{QF}	1.4_{QF}	1_{QF}
FG_3	1.8_{QF}	2.2_{QF}	1.125_{QF}
FG_4	1.8_{QF}	1_{QF}	0.75_{QF}
FG_5	1.125_{QF}	1.4_{QF}	0.875_{QF}

Step 5: A weighted comprehensive decision matrix, depicted in Table 6, can be established by applying the extended multiplication, where QF denotes the quadratic membership function of the form of:

$$2.25_{QF} = 0.3, 1.075, 0.875/2.25/0.1, 1.425, 3.575$$

$$1.8_{QF} = 0.3, 0.875, 0.625/1.8/0.15, 1.375, 3.025$$

$$1.625_{QF} = 0.3, 0.875, 0.45/1.625/0.1, 1.1, 2.625$$

$$1.575_{QF} = 0.2, 0.75, 0.625/1.575/0.5, 2.825, 3.9$$

$$1.4_{QF} = 0.225, 0.625, 0.5/1.4/0.3, 1.925, 3.025$$

$$1_{QF} = 0.15, 0.55, 0.3/1/0.2, 1.3, 2.1$$

$$1.8_{QF} = 0.3, 0.875, 0.625/1.8/0.8, 2.6$$

$$2.2_{QF} = 0.225, 0.975, 1/2.2/0.225, 1.875, 3.85$$

$$1.125_{QF} = 0.3, 0.575, 0.25/1.125/0.05, 0.675, 1.75$$

$$1.8_{QF} = 0.2, 0.85, 0.75/1.8/0.1, 1.225, 2.925$$

$$1_{QF} = 0, 0.375, 0.625/1/0.225, 1.425, 2.2$$

$$0.75_{QF} = 0.15, 0.4, 0.2/0.75/0.1, 0.75, 1.4$$

$$1.125_{QF} = 0.2, 0.55, 0.375/1.125/0, 0.5, 1.625$$

$$1.4_{QF} = 0.15, 0.625, 0.625/1.4/0.15, 1.225, 2.475$$

$$0.875_{QF} = 0.225, 0.45, 0.2/0.875/0.15, 1.025, 1.75$$

Step 6 is to define each sub-goal as a fuzzy number with $i = 1, 2, 3, 4, 5$. The mathematical computations of the 5 sub-goals are in the followings:

$$1.982_{QF} = 0.3, 0.942, 0.65/1.982/0.117, 1.3, 3.075$$

$$1.325_{QF} = 0.192, 0.658, 0.475/1.325/0.333, 2.017, 3008$$

$$1.708_{QF} = 0.275, 0.542, 0.525/1.183/0.142, 1.133, 2.175$$

$$1.183_{QF} = 0.117, 0.542, 0.525/1.183/0.142, 1.133, 2.175$$

$$1.133_{QF} = 0.192, 0.542, 0.4/1.133/0.1, 0.917, 1.95$$

Step 7: Extended average of all sub-goals by means of extended addition and scalar multiplication is in the form of:

$$1.448_{QF} = 0.2152, 0.6984, 0.535/1.448/0.157, 1.297, 2.588.$$

After applying *step 8* and *step 9*, we get the following values:

$$FG_1 = 0.732; FG_2 = 0.579; FG_3 = 0.642; FG_4 = 0.503; \text{ and } FG_5 = 0.444$$

Step 10 is to arrange these values in ascending order and select the alternatives with the maximum ranking value as the best alternative. Therefore, in our case:

$$FG_1 > FG_3 > FG_2 > FG_4 > FG_5.$$

For the validation of the results of the proposed work, we consider an example of Institute Examination Systems. In this example, we identify five functional goals and three non-functional goals; and after applying the proposed method, we identify that FG_1 , i.e., teacher module, is the most important sub-goal and it has the highest priority among all the sub-goals; and FG_5 has the lowest priority. By addressing high priority goal, i.e. FG_1 , before low priority goal, we can significantly reduce the cost and duration of project.

If we compare the results of this paper with other work like AGORA, i^* , KAOS etc., then we identify the following:

In AGORA, preference matrix and contribution values are used for the selection and prioritization of goals. Preference matrices are attached to a goal and stand for the degree of preferences for the goal for each stakeholder. Contribution values are attached to an edge of an AND/OR graph; and it express the degree of contribution of the goal to the achievement of its connected parent goal (Kaiya et al. 2005). In a similar study, Yamamoto and Saeki (2007) applied the AGORA method for the selection of software components. In 2005, Kaiya et al. use the AGORA method for the detection of requirements

discordances among stakeholders. In these studies, crisp data was used for the selection, prioritization, and detection of discordances among stakeholders in requirements elicitation process. Similarly, KAOS, i^* etc. use crisp data for the following activities like selection, prioritization etc. in requirements elicitation process. In practical group requirements elicitation process, stakeholders use linguistic variables like high, low, and very high instead of 1, 2, 3 etc. to specify the degree of preference of goal. Therefore, in-order to strengthen the goal oriented requirements elicitation process like AGORA, i^* , KAOS etc. we proposed a fuzzy based approach for the selection of goals in goal oriented requirements elicitation process by considering a method which consists of ten step, as shown in Fig. 2.

Properly selecting and prioritizing goals provides the following benefits to the project (Firesmith 2004): (i) it improves customer satisfaction by increasing the likelihood that the customer's most important requirements are delivered first (ii) it enables the project manager and customers to modify the project schedule to deal with the project realities of limited resources and fixed deadlines.

5 Conclusion

The problem of selecting goals in requirements elicitation process is a MCDM problem as it includes several stakeholder's participation. Therefore, this paper presents a fuzzy based approach for choosing and adopting a goal out of the alternatives of the goals when multiple stakeholders participate in requirements elicitation and analysis process. In order to make decision when the information is fuzzy and imprecise, in our method, we used α -level weighted F-preference relation in group decision making process.

To simply show how proposed approach works, a numerical example is shown to illustrate the fuzzy group decision making approach in goal oriented requirements elicitation and analysis process. In our example, we assumed that there are five functional goals, i.e., student module (FG_1), teachers module (FG_2), controller of examination module (FG_3), administrative module (FG_4), and online conduct of examination module (FG_5); and three non-functional goals (NFG) or criteria, i.e., effort (NFG_1), cost (NFG_2), risk (NFG_3) and two stakeholders' are participating as decision maker. On the basis of our analysis, we identify that for institute examination systems, FG_1 is the most important sub-goal and it is the best alternative. This goal would be considered in the first release of the software. The future research agenda can be listed as follows:

- (a) To propose a fuzzy attributed goal oriented requirements elicitation and analysis (FAGOREA) method

by using fuzzy contribution values and fuzzy preference matrix using fuzzy analytic hierarchy process (FAHP) and fuzzy TOPSIS.

- (b) To develop a tool to support the decision making processes of FAGOREA.
- (c) To develop a fuzzy goal graph model for functional and non-functional requirements correlation analysis in goal oriented requirements elicitation process.
- (d) To develop a tool to classify the requirements elicited from stakeholders according to the evaluation of their priorities on the basis of software risk and cost.
- (e) To propose a fuzzy goal oriented modelling of requirements elicitation process (FGOMREP) using pattern based approach.

References

- Akker JM, Brinkkemper S, Diepen G, and Versendaal J (2005) Determination of the next release of a software product: an approach using integer linear programming. In: *Proceedings of the CAiSE*, pp. 119–124
- Anton AI (1996) Goal based requirements analysis. In: *IEEE international requirements engineering conference*, pp. 136–144
- Anwer S, Ikram N (2006) Goal oriented requirement engineering: a critical study of techniques. In: *12th Asia pacific software engineering conference*
- Chan L-K, Wu M-L (2007) Quality function deployment: a comprehensive reviews of its concepts and methods. *Qual Eng*, Taylor Francis 15(1):23–35
- Chang PL, Chen YC (1994) A fuzzy multi-criteria decision making method for technology transfer strategy selection in biotechnology. *Fuzzy Sets Syst* 63:131–139
- Chung L, Nixon BA, Yu E, Mylopoulos J (2000) *Non functional requirements in software engineering*. Kluwer Academic Publisher, Massachusetts
- Cochran JK, Chen H (2005) Fuzzy multi—criteria selection of object-oriented simulation software for production system analysis. *Comput Oper Res* 32:153–168
- Firesmith D (2004) Prioritizing requirements. *J Object Technol*. 3 (8):35–47
- Fukushima T, Yamada S (2010) Software process improvements by process monitoring and quality evaluation activities. *Int J Syst Assur Eng Maint* 1(4):284–292
- Hwang CL, Yoon K (1981) *Multiple attributes decision making methods and applications*. Kluwer Academic Press, Springer
- Kaiya H et al. (2002) AGORA: attributed goal oriented requirements analysis. In: *Proceedings of the IEEE joint conference on requirements engineering*
- Kaiya H et al (2005) Improving the detection of requirements discordances among stakeholders. *Requirements Engineering* Springer, New York, pp 289–303
- Karlsson L, Dahlstedt AG, Regnell DJN, Persson A (2007) Requirements engineering challenges in market driven software development—an interview study with practitioners. *Inf Softw Technol* 49:588–604
- Karsak EE (2004) Fuzzy multiple objective programming framework to prioritize design requirements in quality function deployment. *Comput Ind Eng* 47:149–163
- Kavakli E (2002) *Goal oriented requirements engineering: a unifying framework*. requirements engineering. Springer, Münster
- Lamsweerde AV (2001) Goal-oriented requirements engineering: a guided tour. In: *Proceedings of 5th IEEE international symposium on requirements engineering*. pp.249–263
- Lee H-S (2005) On fuzzy preference relation in group decision making. *Int J Comput Math*, Taylor and Francis 82:133–140
- Li RJ (1999) Fuzzy method in group decision making. *Comput Math Appl-Elsevier* 38:91–101
- Li GD, Yamaguchi D, Nagai M (2007) A grey based decision making approach to the supplier selection problem. *Math Comput Modell* 46:573–581
- Lin HY, Hsu PY, Sheen GJ (2007) A fuzzy based decision making procedure for data warehouse system selection. *Expert Syst Appl-Elsevier* 32:939–953
- Liu CH (2010) *A group decision making method with fuzzy set theory and genetic algorithms in quality function deployment. quality and quantity*. Springer, New York, pp 1175–1189
- Mahmood YA, Ahmadi A, Verma AK, Srividya A, Kumar U (2013) Fuzzy fault tree analysis: a review of concept and application. *Int J Syst Assur Eng Maint*, Springer 4(1):19–32
- Nakamura K (1986) Preference relations on a set of fuzzy utilities as a basis for decision making. *Fuzzy Sets Syst* 20:147–162
- Nuseibeh B, Easterbrook S (2002) *Requirements engineering: a roadmap*, In: *Proceedings of the conference on the future of software engineering*, New York
- Oshiro K et al. (2003) Goal-oriented idea generation method for requirements elicitation. In: *11th IEEE international requirements engineering conference*, pp. 1–2
- Pai WC (2006) A quality enhancing software function deployment model. *Inf Syst Manag* 19(3):20–24
- Potts C (1995) Invented requirements and imagined customers: requirements engineering for off-the shelf software. In: *2nd IEEE international symposium on requirements engineering*, pp. 128–130
- Prasad B (1998) Review of QFD and related deployment techniques. *J Manufact Syst* 17(3):221–234
- Sadiq M, Jain SK (2012) An insight into requirements engineering processes. In: *3rd International conference on advances in communication, network, and computing*. LNCSIT-Springer, Chennai, pp. 313–318
- Sadiq M, Jain SK (2013) A fuzzy based approach for requirements prioritization in goal orientation requirements elicitation process. In: *Proceedings of 2013 international conference on software engineering and knowledge engineering*. 27 June 2013
- Sadiq M, Shahid M (2009) Elicitation and prioritization of software requirements. *Int J Recent Trends Eng* 2(3):138–142
- Shibaoka M et al (2007) GOORE: goal oriented and ontology driven requirements elicitation method. *ER Workshop*. LNCS Springer, New York, pp 225–234
- Wang X-T, Xiong W (2011) An integrated linguistic-based group decision making approach for quality function deployment. *Experts Syst Appl* 38:14428–14438
- Yamamoto K, Saeki M (2007) Using attributed goal graphs for software component selection: an application of goal-oriented analysis to decision making. In: *26th international conference on conceptual modeling-ER*
- Yu ESK (1997) Towards modeling and reasoning support for early and late phase of requirements engineering. In: *IEEE international conference on requirements engineering*
- Zadeh LA (1965) Fuzzy sets. *Infor Control* 8:338–353
- Zadeh LA (1975) The concept of a linguistics variable and its application to approximate reasoning (I). *Infor Sci* 8:199–249
- Zhu MX, Luo X-X, Chen XH, Wu DD (2012) A non-functional requirements trade off model in trustworthy software. *Infor Sci-Elsevier* 191:61–75
- Zickert F (2010) Evaluation of the goal oriented requirements engineering methods KAOS. In: *American conference on information system*. pp.1–9